

# The Plan: A Controlled, Text-Aware Market-Neutral Strategy

FIAM-McGill Asset Management Hackathon 2026 · Stage One · Revised plan



**In one sentence.** A moderately good, honestly validated signal, combined through a regime-aware gate and delivered through a beta-, sector- and dollar-neutral book at full gross, will score better on this mandate than a more exotic model with a noisier book.

## 1. The principle behind every choice

You are scored mainly on the information ratio against the 3-month T-bill plus 4%. For a dollar-neutral book whose collateral earns the bill rate:

$$IR \approx \text{Sharpe}(\text{spread}) - \sqrt{12} \times (0.04/12) / (k \times \sigma_{\text{spread}})$$

where  $k$  is gross exposure and  $\sigma_{\text{spread}}$  is the volatility of the long-short spread. This implies three things:

- The 4% hurdle is a fixed drag, so **run the full 200% gross**.
- **Unrewarded risk costs as much as missing alpha**, so neutralize beta, sectors and size tightly.
- The committee checks neutrality before reading anything else.

The target is therefore a moderately good signal, delivered through a very clean book.

## 2. Data and preprocessing

- **Target.** *ret\_exc\_lead1m*, the excess return of month  $t+1$ . Map it to *stock\_exret* and date each row by its target month, keeping the original *eom* alongside. Never shift it again, and never use it as a predictor.
- **Characteristics.** Rank-transform all 147 to  $[-1, 1]$  within each month. Fill missing values with the monthly median after ranking, and add missing-value flags for the most frequently missing variables.
- **Universe.** Each month, exclude the bottom 20% by market equity and stocks priced under about \$5, using only month- $t$  information. Define the prediction universe from what is known at  $t$ , never from whether next month's return exists, as the brief explicitly requires.
- **Market data.** TB3MS from FRED for the benchmark, and S&P 500 returns for beta and alpha. FRED's S&P 500 series is a price index without dividends; if you use it, say so, or use a total-return source.

## 3. Signals

### Characteristic groups

Split the 147 characteristics into four specialists:

- value and fundamentals
- momentum and past returns
- quality, profitability and accruals
- liquidity and risk

### Text features from 8-K filings

Use only filings whose *filing\_date* falls inside month *t*.

- FinBERT tone scores (positive, negative, neutral), aggregated per stock-month, computed on text with company names and tickers removed.
- Item-code indicators, especially 4.02, 5.02, 2.06 and 1.02, plus filing counts.
- A “no filing this month” flag, treated as a state rather than as missing data.
- Optionally, FinBERT embeddings reduced to 10-20 components with PCA fitted on training data only.

These form a fifth specialist. FinBERT’s training text predates the 2021 test window, which is your answer to model-side look-ahead.

### Market state variables

The same for every stock in month *t*:

- cross-sectional return dispersion
- trailing average pairwise correlation
- average idiosyncratic volatility
- trailing market return and volatility

## 4. The model

- **Specialists.** Train one shallow, strongly regularized LightGBM per group. The target is next-month return ranked or demeaned within the month, so the models learn relative performance rather than market direction.
- **Gate.** Combine the five specialists with weights that depend on the market state variables. Keep it simple, for example a ridge regression on specialist forecasts and their interactions with one or two state variables. Train the gate on the validation window, where the specialists’ forecasts are genuinely out of sample. This is your regime awareness: learned, testable and plottable.
- **Baselines.** Run the template’s OLS, Ridge, Lasso and Elastic Net on all characteristics, plus a single LightGBM on everything. These show what the specialist-and-gate structure adds.

- **Schedule.** Use an expanding training window and a rolling two-year validation window, assigned by target month, with annual refits:

| Test year  | Training        | Validation |
|------------|-----------------|------------|
| 2021       | 02/2015–12/2018 | 2019–2020  |
| 2022       | through 2019    | 2020–2021  |
| ...        | ...             | ...        |
| 01-08/2026 | through 2023    | 2024–2025  |

- **OOS  $R^2$** , using the brief's zero-benchmark formula, for every model.

## 5. Portfolio construction

- **Smoothing.** Apply an exponential moving average over two to three months to the forecasts before optimizing, which cuts turnover.
- **Monthly optimization (cvxpy).** Maximize forecast exposure subject to:

| Constraint     | Setting                                    |
|----------------|--------------------------------------------|
| Positions      | About 150 long and 150 short               |
| Gross exposure | 200%                                       |
| Net exposure   | $\approx 0$ (far inside $\pm 50\%$ )       |
| Beta           | $\approx 0$ , using shrunk <i>beta_60m</i> |
| Sectors        | Exposures $\approx 0$                      |
| Single name    | $\leq 1.5\%$                               |
| Turnover       | Penalized in the objective                 |

- **Hold buffer.** Enter at the top or bottom 10% of forecasts; exit only when a stock leaves the top or bottom 20%.
- **Capital convention.** \$100 of capital, \$100 long and \$100 short, with the collateral earning the bill rate. Total return equals the cash return plus the long-short return. Active return equals that total minus TB3MS/100/12 minus 0.04/12. State this explicitly in the deck.
- **Optional volatility-scaled gross exposure.** Adopt it only if validation shows your signal weakens when volatility is high.

## 6. Evaluation

- Every statistic on the brief's Page 4–7 checklist, for the strategy, benchmark and S&P 500: returns, IR, Sharpe, alpha and beta with Newey-West standard errors, correlation, drawdowns, turnover, exposures, concentration, and short-book size and liquidity.
- The required charts, including rolling 12-month beta and the top 10 positive and negative contributors by ticker and company name.

- **Regime breakdown:** 2021 rebound, 2022 drawdown, 2023–2025 concentration, 2026. Show which specialist and which leg worked in each.
- **Ablations:** with and without text; gate versus equal weighting; specialists versus a single model.
- A plot of the gate weights over time, as your explainability exhibit.

## 7. Integrity

- Scripted leakage checks: date-alignment assertions, a check that no feature uses month  $t+1$  data, and a shuffled-label test that should produce roughly zero performance.
- Lock all design choices on validation data before looking at 2021–2026 results.
- Keep a research log of everything tried, including failures, for the appendix.

## 8. Deliverables

| Item             | Content                                                                                         |
|------------------|-------------------------------------------------------------------------------------------------|
| <b>Page 1</b>    | Executive summary: strategy, IR against cash plus 4%, beta, S&P 500 for context                 |
| <b>Page 2</b>    | Construction, neutrality definition, signals, top 10 longs and shorts, cumulative returns chart |
| <b>Page 3</b>    | Data, methodology, why an encoder, training schedule, OOS $R^2$ table                           |
| <b>Pages 4–7</b> | Performance pack; overflow goes to the appendix                                                 |
| <b>Page 8</b>    | Discussion: what worked, in which regime, key positions, improvements                           |
| <b>Files</b>     | Holdings CSV, returns CSV, <i>MAIN.py</i> , CVs                                                 |

Verify every ticker and company name label against the permno-dated links, and record the source for any unlabeled holding.

## 9. Order of work

If the deadline follows prior years (early October), you have about a week. Build in this order, so that each step leaves you with a submittable result:

- 1 Data pipeline, baselines and the neutral optimizer.
- 2 Specialists and gate.
- 3 FinBERT text features and the ablation.
- 4 Full evaluation pack and deck.
- 5 Optional extras, such as volatility-scaled gross or daily risk measurement, only if time remains.